

Adeola Damilola Kola-Olaleye

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EDUCATION

Nile University of Nigeria

Abuja, Nigeria

Bachelor of Science (Hons) in Computer Science; 4.42/5.00 GPA

2024 - 2026

- **Relevant Coursework:** Operations Research, Numerical Analysis, Electronics Systems, System Design and Analysis.
- **Thesis:** Cognitive Vision-Based Hand Gesture Recognition System for Media and Accessibility Control.

University of Pécs

Pécs, Hungary

Bachelor of Science in Computer Science (Transferred); 4.24/5.00 GPA

2019 - 2023

- Software Engineering Concentration.

RESEARCH INTERESTS

Core Focus: Touchless Assistive Interfaces, Cognitive Vision for Frustration Detection, Age-Adapted Gesture Recognition Systems, Hardware Prototyping, and Signal Measurement.

RESEARCH EXPERIENCE

Cognitive Vision Hand Gesture Recognition System

Abuja, Nigeria

Undergraduate Thesis / Final Year Project

2025

- Engineered a real-time, touchless HCI system using Python, OpenCV, and Google MediaPipe to map bi-manual 2.5D hand landmarks to continuous OS-level controls (WASAPI/DDC-CI) with an average latency of 35 ms; 28-30 FPS in good lighting, 15-20 FPS in low light on consumer-grade CPUs.
- Implemented geometric regression algorithms, linear interpolation, and palm-depth proxies (Z-axis) alongside a gesture clutch mechanism to eliminate false activations and tremor-induced signal jitter.
- Selected to showcase the system at the faculty thesis exhibition panel.

PROJECTS

ROS Robot - Controller Input: Configured teleoperation nodes and subscriber/publisher communication pipelines for a chassis robot on Ubuntu OS and ROS Noetic. Integrated Arduino microcontrollers and utilized digital multimeters and logic analyzers for precise signal measurement and hardware prototyping. Tech: Raspberry Pi 3 Model B, Arduino, CamJam EduKit, ROS Noetic, Python.

Pathfinding Visualization GUI: Implemented an interactive Pygame visualizer for graph traversal and search algorithms. Supports execution of A* Search, Dijkstra's Algorithm, Breadth-First Search (BFS), and Depth-First Search (DFS) to dynamically identify optimal trajectories across grid obstacles.

Not3s - Cross-Platform Note-Taking Application: Developed a seamless cross-platform note-taking application leveraging Dart and Flutter, featuring intuitive UI/UX design and real-time backend data synchronization.

PROFESSIONAL EXPERIENCE

Automation Developer Intern

Budapest, Hungary

ExxonMobil

2021 - 2022

- Developed and maintained workflow automation scripts aligned with enterprise CI/CD and standard documentation frameworks.
- Collaborated in an Agile DevOps environment to identify and eliminate manual pipeline bottlenecks across cross-functional teams, reducing overall deployment delays.

Software Developer Intern

Lagos, Nigeria

WebSoft Inc

2018 - 2019

- Engineered responsive web modules and backend components using clean code architectures and standard version control (Git).
- Authored technical API documentation and system specifications to streamline feature handoffs across the development team. Conducted code reviews, unit testing, and debugging to maintain platform stability and improve overall load performance.
- Interfaced with clients and senior developers during sprint planning sessions.

Mid-Semester Summer Instructor

Pécs, Hungary

Summer Camp, University of Pécs

2019 - 2023

- Developed and delivered an introductory programming curriculum using MS Excel and VBA to automate real-world data processing workflows.
- Taught and mentored students in preparation for the TOEFL exam and technical entrance assessments for European STEM degree programs.

SKILLS

Programming: Python, Dart, C++.

Technical: Firebase, MySQL, OpenCV, TensorFlow, ROS, Arduino, GPIO Interfacing.

Version Control & Tools: Git/GitHub, Linux, $\text{\LaTeX}$.

Languages: English (Native), Hungarian (Elementary).

AWARDS

Dean's Merit Award

University of Pécs

Pécs, Hungary

2021 - 2022 & 2022 - 2023